



Gene Set Enrichment Analysis in Whole Body Hyperthermia Treatment





Roadmap

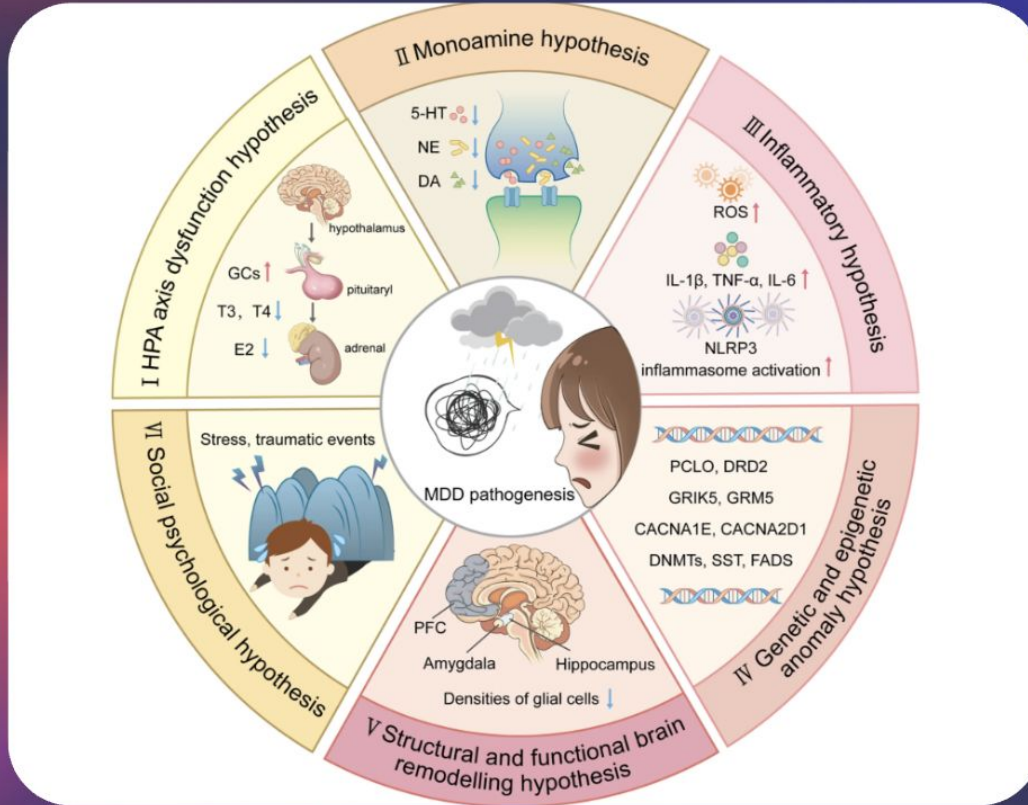




Biological Background:

WBH As a Potential Treatment for Depression



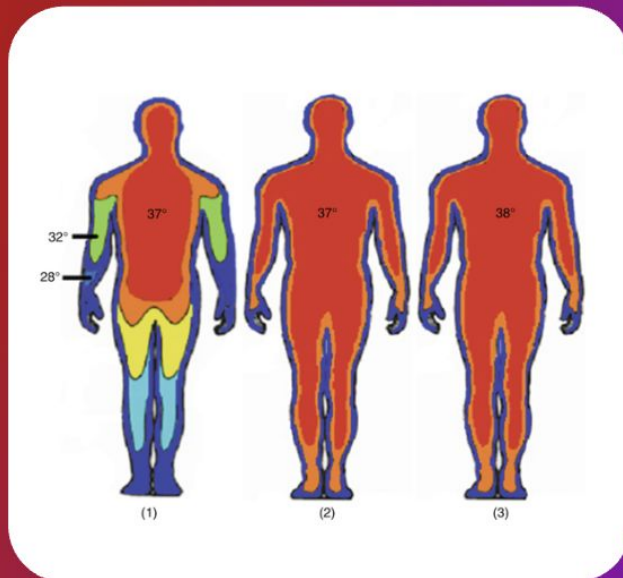


(Cui et al., 2024)

Biological Motivation

- **Whole-body hyperthermia (WBH)** is a medical procedure that intentionally raises body temperature to high heat ($\sim 38.5^{\circ}\text{C}$)
- **Heat exposure** activates heat shock proteins and immune pathways, which recent research suggests might help **treat depression**

(Akonom et al., 2026)



(Heckel-Reusser, 2022)

How Does GSEA Help?

POTENTIAL EXPLORATORY PATHWAYS



Cellular Response to Heat

Chaperone Mediated Protein Folding

Dopamine Transport*

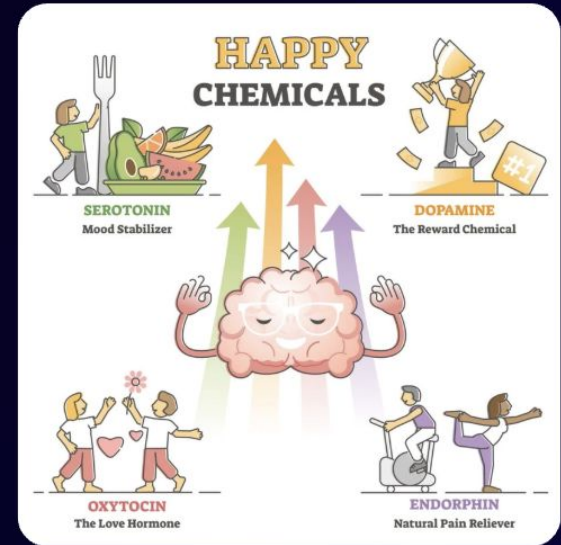
Serotonin Transport*

*indicates novel relationships not yet explored in paper

- The effects of heat exposure on **individual genes** in WBH are very **small**
- However, by analyzing groups of **related genes**/biological pathways, a significant pattern might appear
- This matches the context in which **GSEA** is typically used - detecting **small expression fold changes across many genes**

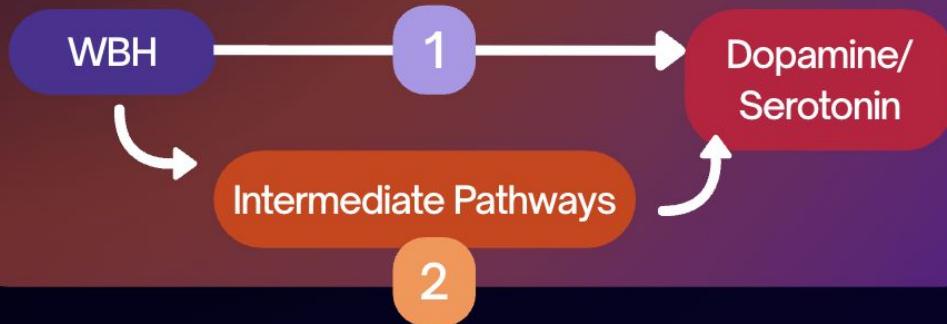
- Akonom and colleagues identified heat shock, immune response, and interleukin 6 production pathways as enriched in the WBH condition
- But how might this affect **mood-relevant targets** like **dopamine** or **serotonin** → so-called “happy chemicals”?
- **Major Depressive Disorder** is linked to **deficits in dopamine and serotonin circuits**

Focus Pathways

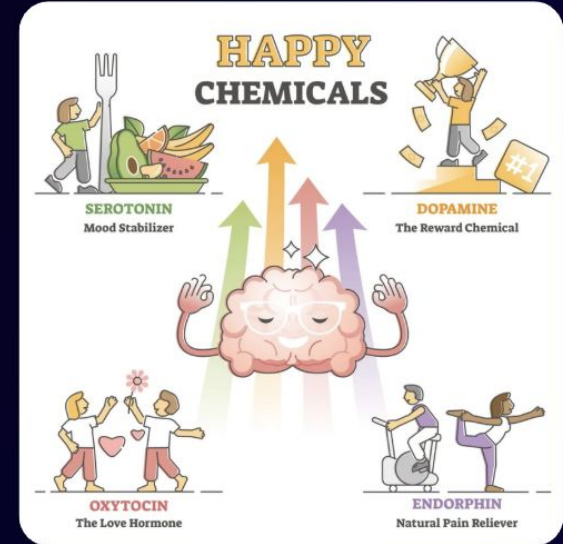


(Simply Psychology)

- If WBH helps treat depression, we expect
 - 1. Serotonin and dopamine pathways are **themselves enriched** in WBH condition
- OR ...
 - 2. **Indirectly benefit** from the robust associations of heat shock, protein folding



Focus Pathways



(Simply Psychology)

Ethan Aidam, Victoria Chen, Anisha Kolambe



Reviewing GSEA: Enrichment Scores and Statistical Justification



What is



?

- GSEA is a **computational tool** that identifies if a set of genes is overrepresented in a biological dataset
- Calculating “**enrichment scores**” helps determine whether the **gene** is **linked** to a known **biological pathway**

RANK GENES BASED ON LOG2FOLD CHANGE



CALCULATE ENRICHMENT SCORE



COMPUTE P-VALUES W/ PERMUTATION TEST



PLOT AND INTERPRET DATA



W



Enrichment Score (ES)

Consider a list of NR ranked genes, $L = \{g_1, g_2, \dots, g_N\}$, and an independent gene set, S , consisting of N_H genes.

For each position $i = 1, 2, \dots, N$ we compute the fraction of genes that are in S (call these “hits”) and the fraction of genes that aren’t in S (call these “misses”)

$$P_{hit}(S, i) = \sum_{g_j \in S, j \leq i} \frac{|r_j|^p}{N_R} \quad \text{where,} \quad N_R = \sum_{g \in S} |r_j|^p$$

$$P_{hit}(S, i) = \sum_{g_j \notin S, j \leq i} \frac{1}{N - N_H}$$

An “enrichment score” is defined as follows:

$$ES(S) = \max(|P_{hit} - P_{miss}|)$$



Significance Testing

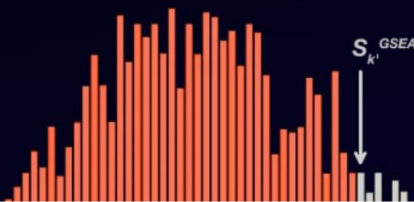


Phenotype Permutation

To assess significance, the observed ES is compared to a null distribution ES_{NULL} with random phenotypes.

1. Randomly **shuffle** phenotypes and **recompute** the ES
2. Repeat this 1,000 times to **generate a distribution** of null scores (ES_{null}).
3. **Calculate** the p-value from the fraction of values greater or equal to the observed ES within the null distribution

(Subramanian et al., 2005)



Normalized Enrichment Score (NES)

Normalized Enrichment Scores (NES)

allows us to compare ES across gene sets of different sizes.

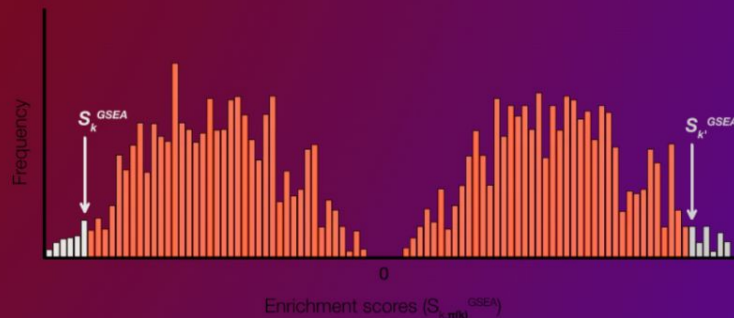
We normalize each ES by dividing by the expected value of the corresponding null distribution.

$$NES(S) = \frac{ES(S)}{\mathbb{E}[ES_{null}(S)]}$$

More On Null Distributions

ES is **positive** for enrichment at the **top of the ranked list** and **negative** for enrichment at the **bottom of the list**, which produces two different null distributions for each set S . Thus, for Normalized Enrichment Scores (NES) and p-values, we derive them from the null distribution corresponding with the sign of the observed ES for each set.

$$NES(S) = \begin{cases} \frac{ES(S)}{\mathbb{E}[ES_{null}^+]} : ES(S) > 0 \\ \frac{ES(S)}{\mathbb{E}[ES_{null}^-]} : ES(S) < 0 \end{cases}$$





Multiple Testing Correction:

Benjamini Hochberg (BH) + Storey's Method



Why Do We Need Correction?

- In our paper, researchers are simultaneously **testing thousands of genes** and biological **pathways** for enrichment
- If we used the **standard threshold $p < 0.05$** , it would be easy for about 5% of tests to be **significant purely by chance**, because so many were examined

Ex.

Testing **20,000 genes** ->

- $20,000 \times 0.05 = \mathbf{1,000}$ **expected false positives** by chance alone
- This leads to an inflated **false discovery rate** that needs correcting through BH or Storey's

$$FDR = \text{False Positives} / \text{Total Significant Results}$$

Benjamini-Hochberg

vs.

Storey's Method

1. **Rank** p-values from smallest $\rightarrow$ largest
2. For each rank, **compute** $k\alpha/m$ where k = rank, α = desired FDR, and m = # tests
3. Find the **largest p-value** that is less than or equal to $k\alpha/m$, everything below that passes!

*The paper found that pathways with low FDR were more reliable signals related to WBH treatment effects.

1. Unlike BH, where $m_0 = m$, **estimate** π_0 to be m_0/m , the **proportion** of truly null tests
2. Then, the FDR estimate is **adjusted** accordingly to get $k\alpha/m\pi_0$ as the threshold
3. If **many pathways** show signal, $\pi_0 < 1$

*Storey's is less conservative than BH, which in turn is less conservative than FWER tests like Bonferroni

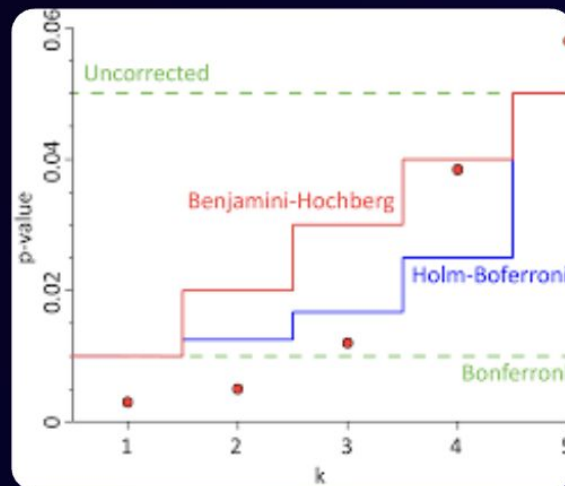
FDR-adjusted Analysis ($q < 0.05$)

- **88** significantly enriched biological processes
- E.g., Heat shock, immune signalling, etc.
- **Higher sensitivity**/less strict method of control

FWER-adjusted Analysis ($p < 0.05$)

- Only **17** significantly enriched processes
- **Strongest**: protein folding, cellular heat response
- **Conservative**, retains most robust associations

FDR vs. FWER, Paper Findings



(Marek Gierliński)



Paper Methodology:

Cutadapt, HISTA2, Rsubread, NCBI Annotation



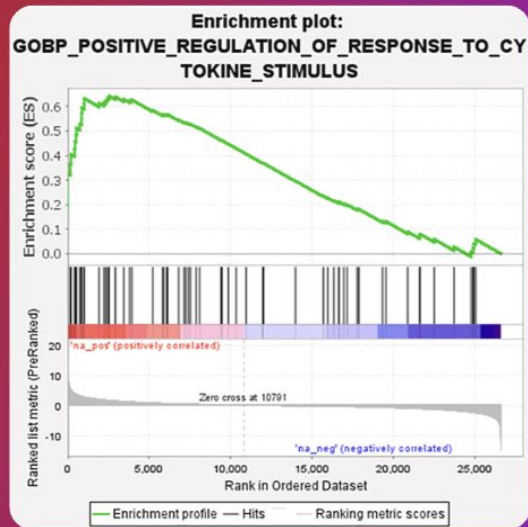
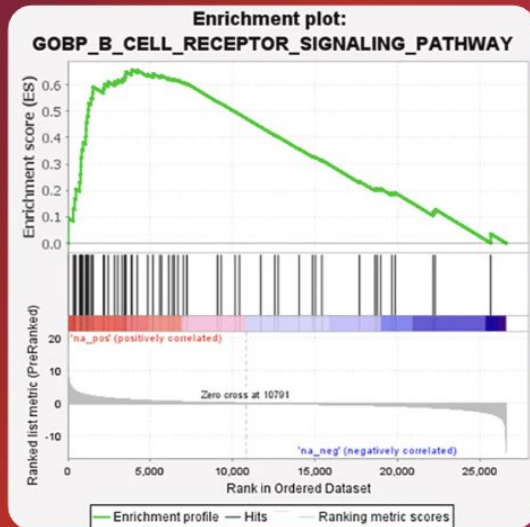
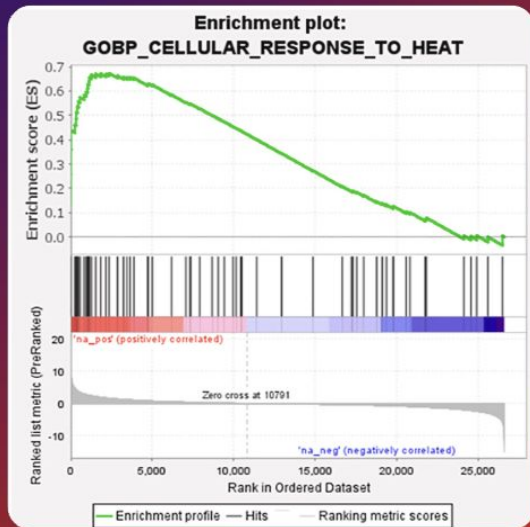
Paper Methodology

- Trimmed Reads using using Cutadapt (v2.1)
- NCBI GRCh38 alignment via HISAT2 (v2.1.0)
- Counts generated by Rsubread (v2.0.1)
- NCBI RefSeq annotation
- RRCov with PCA
- DESeq2 normalization, DEG analysis, and ranking
- 88 BPs upregulated

(Akonom et al., 2026)

GO Biological Process	SIZE	ES	NES	FWER-adjusted p-value
Chaperone mediated protein folding	68	0.7681	2.629	<0.001
Chaperone cofactor dependent protein refolding	32	0.8310	2.522	<0.001
De novo protein folding	41	0.7933	2.498	<0.001
Protein folding	214	0.6031	2.441	<0.001
Protein refolding	22	0.8313	2.313	<0.001
Chaperone mediated protein complex assembly	22	0.8141	2.283	<0.001
Cellular response to heat	63	0.6733	2.260	0.002
Ribonucleoprotein complex subunit organization	228	0.5514	2.237	0.003
Positive regulation of response to cytokine stimulus	64	0.6429	2.210	0.004
B cell receptor signaling pathway	62	0.6567	2.206	0.004
Positive regulation of telomerase RNA localization to Cajal body	15	0.8706	2.192	0.006
Peptide antigen assembly with MHC protein complex	18	0.8208	2.126	0.018
Response to topologically incorrect protein	160	0.5409	2.123	0.022
Positive regulation of type I interferon production	60	0.6195	2.112	0.03
Ribonucleoprotein complex biogenesis	461	0.4880	2.108	0.031
Positive regulation of interferon beta production	37	0.6899	2.099	0.033
Inclusion body assembly	25	0.7509	2.083	0.043

GSEA Results



(Akonon et al., 2026)

GSEA Results (App)

Enrichment in phenotype: **na**

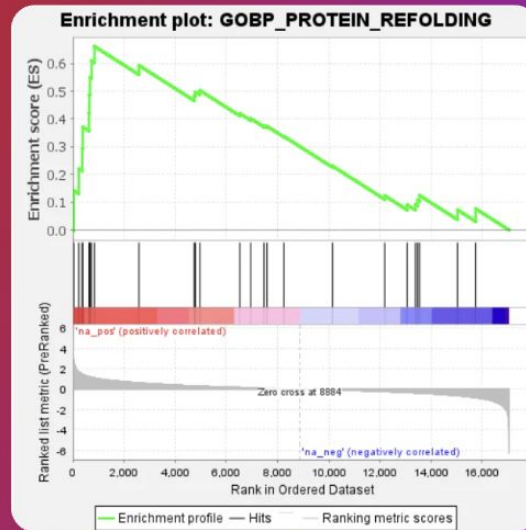
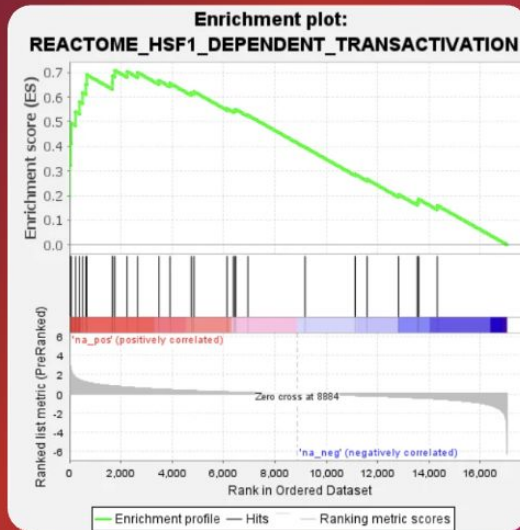
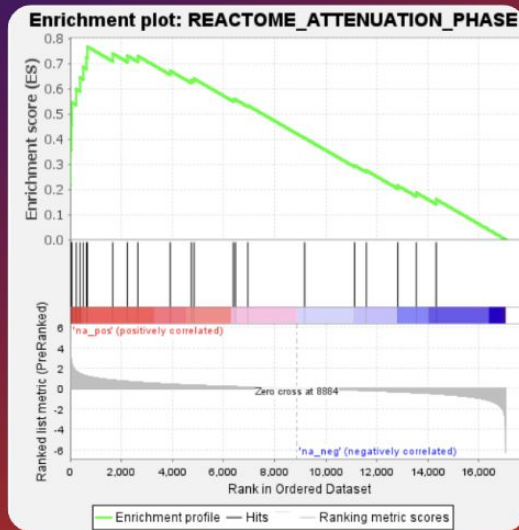
- 1968 / 4735 gene sets are upregulated in phenotype **na_pos**
- 9 gene sets are significant at FDR < 25%
- 36 gene sets are significantly enriched at nominal pvalue < 1%
- 132 gene sets are significantly enriched at nominal pvalue < 5%
- [Snapshot](#) of enrichment results
- Detailed [enrichment results in html](#) format
- Detailed [enrichment results in TSV](#) format (tab delimited text)
- [Guide to](#) interpret results

Enrichment in phenotype: **na**

- 2767 / 4735 gene sets are upregulated in phenotype **na_neg**
- 67 gene sets are significantly enriched at FDR < 25%
- 157 gene sets are significantly enriched at nominal pvalue < 1%
- 392 gene sets are significantly enriched at nominal pvalue < 5%
- [Snapshot](#) of enrichment results
- Detailed [enrichment results in html](#) format
- Detailed [enrichment results in TSV](#) format (tab delimited text)
- [Guide to](#) interpret results

		SIZE	ES	NES	NOM p-val	FDR q-val	FWER p-val	
71	GOBP_CELLULAR_RESPONSE_TO_HEAT	59	0.43	1.50	0.021	0.659	1.000	
37	GOBP_B_CELL_RECEPTOR_SIGNALING_PATHWAY	72	0.45	1.62	0.000	0.469	1.000	
135	GOBP_POSITIVE_REGULATION_OF_RESPONSE_TO_CYTOKINE_	63	0.40	1.41	0.037	0.700	1.000	
1	REACTOME_ATTENUATION_PHASE	Details ...	24	0.77	2.21	0.000	0.001	0.002
2	REACTOME_HSF1_DEPENDENT_TRANSACTIVATION	Details ...	30	0.71	2.16	0.000	0.002	0.005
3	GOBP_PROTEIN_REFOLDING	Details ...	26	0.66	1.94	0.000	0.106	0.345
4	REACTOME_HSP90_CHAPERONE_CYCLE_FOR_STEROID_HORMONE_RECEPTORS_SHR_IN_THE_PRESENCE_OF_LIG	Details ...	49	0.57	1.91	0.000	0.130	0.500
5	GOBP_POSITIVE_REGULATION_OF_SYNAPTIC_TRANSMISSION_GLUTAMATERGIC	Details ...	16	0.72	1.91	0.002	0.111	0.524
6	REACTOME_HSF1_ACTIVATION	Details ...	27	0.65	1.90	0.000	0.102	0.554
7	REACTOME_MITOCHONDRIAL_UNFOLDED_PROTEIN_RESPONSE_UPRMT	Details ...	17	0.70	1.87	0.000	0.143	0.728
8	REACTOME_CD22_MEDIATED_BCR_REGULATION	Details ...	49	0.55	1.85	0.002	0.159	0.816
9	REACTOME_ANTIGEN_ACTIVATES_B_CELL_RECEPTOR_BCR_LEADING_TO_GENERATION_OF_SECOND_MESSENGER	Details ...	74	0.51	1.84	0.000	0.153	0.842

GSEA Results (DESeq2)



GSEA Results (EdgeR)

Enrichment in phenotype: **na**

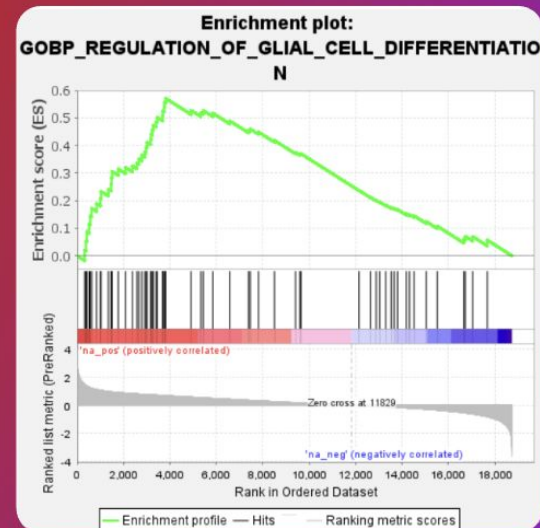
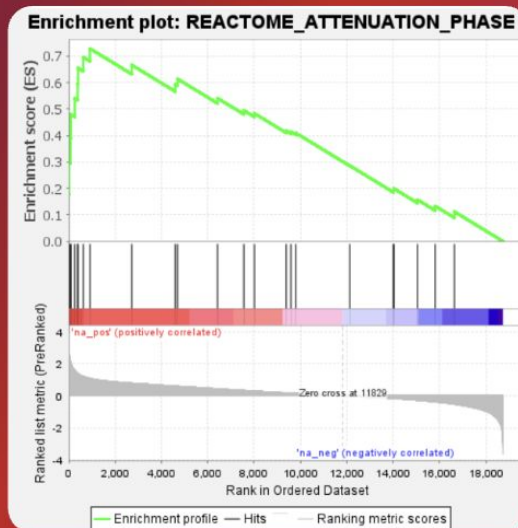
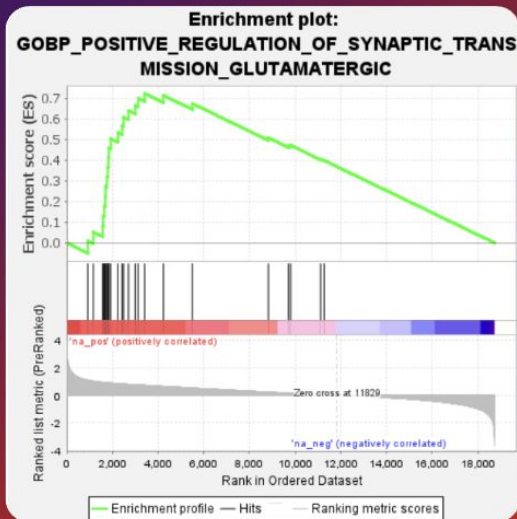
- 3108 / 4996 gene sets are upregulated in phenotype **na_pos**
- 357 gene sets are significant at FDR < 25%
- 246 gene sets are significantly enriched at nominal pvalue < 1%
- 502 gene sets are significantly enriched at nominal pvalue < 5%
- [Snapshot](#) of enrichment results
- Detailed [enrichment results in html](#) format
- Detailed [enrichment results in TSV](#) format (tab delimited text)
- [Guide to](#) interpret results

Enrichment in phenotype: **na**

- 1888 / 4996 gene sets are upregulated in phenotype **na_neg**
- 335 gene sets are significant at FDR < 25%
- 244 gene sets are significantly enriched at nominal pvalue < 1%
- 474 gene sets are significantly enriched at nominal pvalue < 5%
- [Snapshot](#) of enrichment results
- Detailed [enrichment results in html](#) format
- Detailed [enrichment results in TSV](#) format (tab delimited text)
- [Guide to](#) interpret results

		SIZE	ES	NES	NOM p-val	FDR q-val	FWER p-val
379	GOBP_CELLULAR_RESPONSE_TO_HEAT	61	0.42	1.45	0.021	0.264	1.000
228	GOBP_B_CELL_RECEPTOR_SIGNALING_PATHWAY	74	0.43	1.56	0.005	0.161	1.000
923	GOBP_POSITIVE_REGULATION_OF_RESPONSE_TO_CYTOKINE	65	0.34	1.21	0.166	0.574	1.000
1	GOBP_POSITIVE_REGULATION_OF_SYNAPTIC_TRANSMISSION_GLUTAMATERGIC	26	0.72	2.19	0.000	0.002	0.002
2	REACTOME_ATTENUATION_PHASE	24	0.73	2.16	0.000	0.001	0.003
3	GOBP_REGULATION_OF_GLIAL_CELL_DIFFERENTIATION	62	0.57	2.03	0.000	0.012	0.046
4	REACTOME_POTASSIUM_CHANNELS	84	0.55	2.03	0.000	0.009	0.047
5	GOBP_NEURON_FATE_COMMITMENT	48	0.59	2.01	0.000	0.011	0.070
6	REACTOME_NEUREXINS_AND_NEUROLIGINS	50	0.58	1.99	0.000	0.014	0.107
7	REACTOME_CD22_MEDIATED_BCR_REGULATION	49	0.57	1.97	0.000	0.018	0.151
8	GOBP_NEUROMUSCULAR_PROCESS_CONTROLLING_BALANCE	51	0.57	1.97	0.000	0.016	0.155
9	REACTOME_PROTEIN_PROTEIN_INTERACTIONS_AT_SYNAPSES	71	0.54	1.96	0.000	0.016	0.174

GSEA Results (EdgeR)



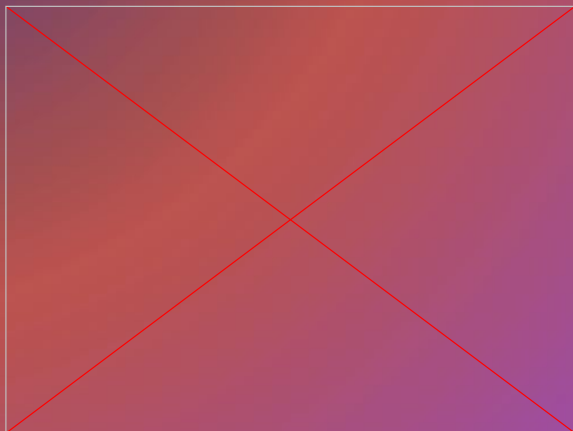


Our GSEA Pipeline:

Shell Scripts, Data Cleaning, R Plots + Analysis



I. Data Organization



LOAD RAW STAR COUNT MATRIX



ASSIGN SAMPLES TO SHAM OR WBH

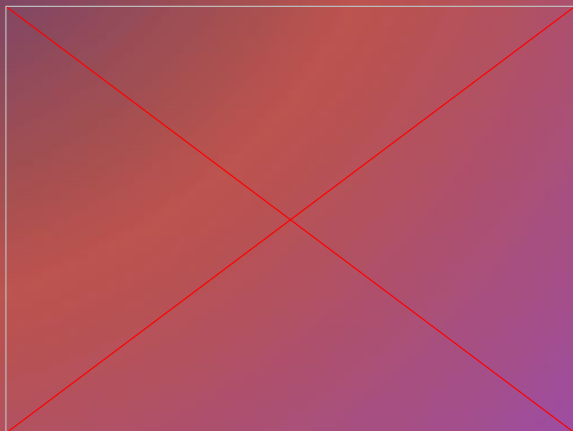


FILTER BY CPM + RUN DESEQ2



CREATE GENE LIST RANKED BY FOLD CHANGE

II. GSEA Pipeline



DEFINE GENE SETS FOR BIO PATHWAYS



CREATE ENRICHMENT SCORE FUNCTION



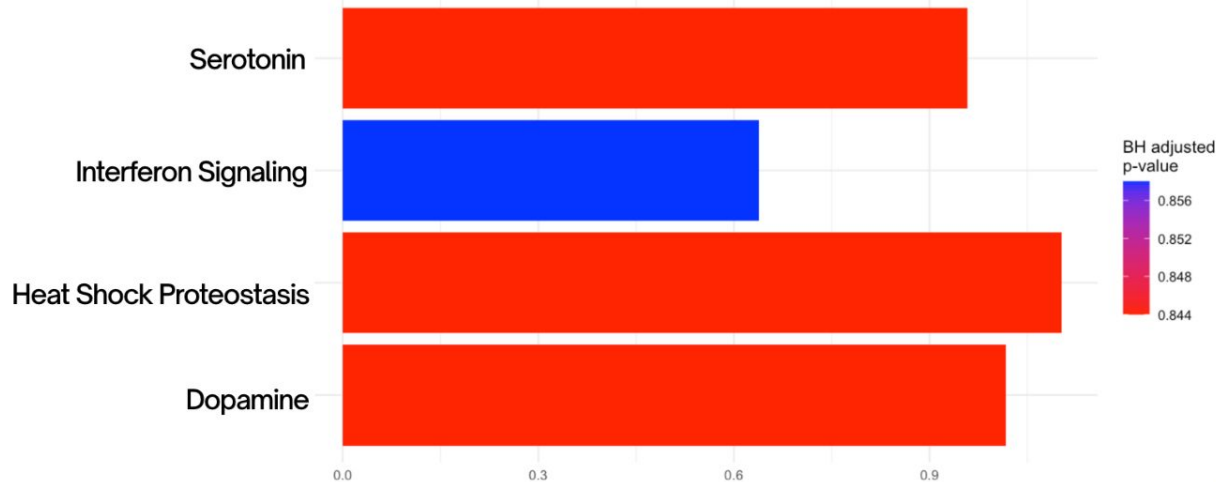
RUN PERMUTATION TEST



PLOT + ANALYZE GSEA RESULTS

Results

NES + Summary Table



Gene Set Enrichment Analysis Results

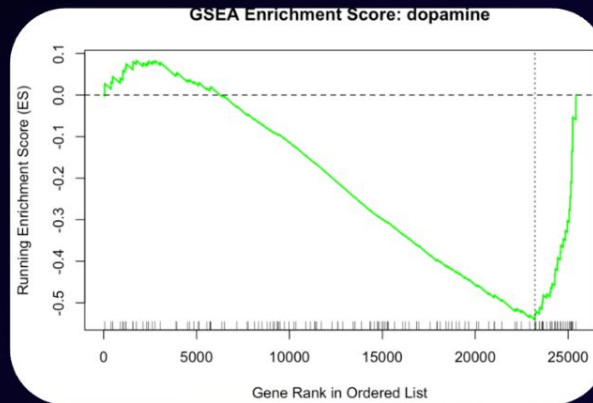
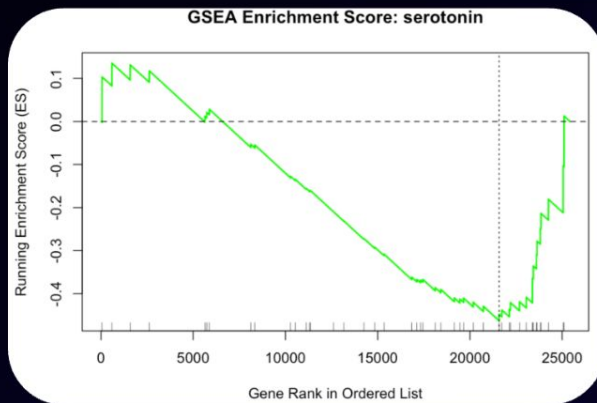
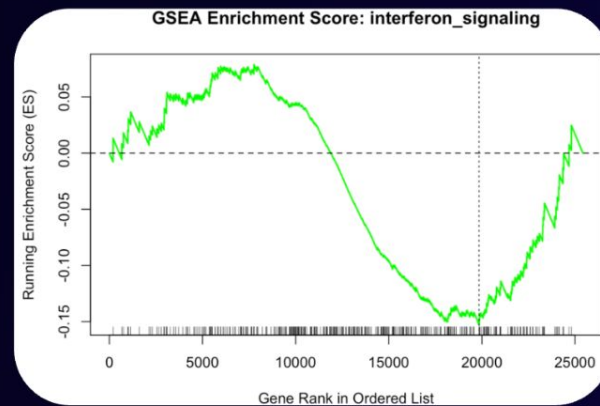
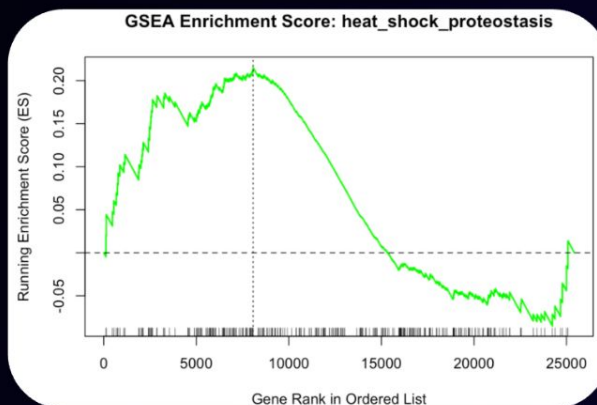
Summary of enrichment scores and significance metrics

Gene Set	Enrichment Score	Normalized ES	P-value	Adjusted P (BH)	Pi 0 (Storey)	Q-value (Storey)
Heat Shock Proteasis	0.215	1.103	0.307	0.844	1	0.307
Dopamine	-0.541	-1.017	0.527	0.844	1	0.527
Serotonin	-0.463	-0.958	0.633	0.844	1	0.633
Interferon Signaling	-0.153	-0.638	0.858	0.858	1	0.858

GSEA Results
by Gene Set

Results

Enrichment Plots



Heat Shock Proteostasis shows a large **early peak** while Interferon Signaling, Serotonin, and Dopamine show **late valleys** after initial shallow peaks which is reflected in their final enrichment scores.

References

Akonom, T. N., Allen, M. A., Begay, T. K., Mason, A. E., Cole, S. P., Raison, C. L., & Lowry, C. A. (2026). Whole-blood transcriptomic response to whole-body hyperthermia in participants with major depressive disorder. *Brain, Behavior, & Immunity - Health*, 54, 101225. <https://doi.org/10.1016/j.bbih.2026.101225>

Cui, L., Li, S., Wang, S. et al. Major depressive disorder: hypothesis, mechanism, prevention and treatment. *Sig Transduct Target Ther* 9, 30 (2024). <https://doi.org/10.1038/s41392-024-01738-y>

Heckel-Reusser, S. (2022). Whole-body hyperthermia (Wbh): Historical aspects, current use, and future perspectives. In P. Vaupel (Ed.), *Water-filtered Infrared A (wIRA) Irradiation* (pp. 143–154). Springer International Publishing. https://doi.org/10.1007/978-3-030-92880-3_11

Subramanian, A., Tamayo, P., Mootha, V. K., Mukherjee, S., Ebert, B. L., Gillette, M. A., Paulovich, A., Pomeroy, S. L., Golub, T. R., Lander, E. S., & Mesirov, J. P. (2005). Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles. *Proceedings of the National Academy of Sciences*, 102(43), 15545–15550. <https://doi.org/10.1073/pnas.0506580102>